

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

Anonymous authors
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ABSTRACT

Counterfactual explanations (CE) characterize the minimal perturbation of an input that changes a model’s prediction; in practice, however, many CE generators return perturbations that modify more coordinates of the feature vector than are strictly required, which degrades interpretability and actionability. We introduce *COLA*, a post-hoc refinement framework that is agnostic to both the predictive model and the CE generator. Given any candidate CE, *COLA* (i) computes an optimal-transport (OT) coupling between the factual set and the counterfactual set, and (ii) uses this coupling to condition a Shapley-value attribution operator (*p*-SHAP) that selects a minimal-cardinality subset of feature edits while preserving the induced target effect. On the theoretical side, we establish that the OT coupling minimizes an upper bound on the 1-Wasserstein (W_1) divergence between factual and counterfactual model outputs, and that, under a Lipschitz-continuity assumption on the model, the refined counterfactuals are provably no farther from the factuals than the input CE. Empirically, across four datasets, twelve predictive models, and five CE generators, *COLA* attains identical target effects while retaining only 26–45% of the feature edits present in the original CE. On a tractable small-scale benchmark, *COLA* closely matches the exact optimum.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) provides the methodological basis for rendering artificial intelligence systems transparent and trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Within this literature, feature attributions (FA) methods—prominently Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017)—quantify the contribution of each input feature to the output of a machine learning (ML) model, thereby reducing the effective dimensionality of an explanation to the most influential coordinates. In a clinical predictive model, for instance, Shapley values isolate the dominant covariates, such as age and medical history, and thereby support clinician interpretation of the model’s decisions (Ter-Minassian et al., 2023; Nohara et al., 2022). Counterfactual Explanations (CE) (Wachter et al., 2017; Guidotti, 2022) constitute a complementary technique that quantifies how localized perturbations of the input map to a change in the model output. Although the literature contains hundreds of CE algorithms (Guidotti, 2022; Verma et al., 2020), no single algorithm is universally applicable, since each is specialized to a particular problem setting, objective, and task. The objective specification varies across methods: some formulate the task as recovering a single counterfactual instance per factual instance, whereas others treat a set of instances jointly and seek one or several counterfactual instances per factual observation. A further class of methods operates at the dataset level, computing global CE that specify a shared displacement direction driving the factual instances toward the desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024). Yet other formulations model the set of factual instances as an empirical distribution and seek a counterfactual distribution that is close in shape to the factual distribution (You et al., 2025) while incurring comparable cost. The methods also differ in their model assumptions: some presuppose differentiability of the predictor, whereas others are constructed specifically for tree-based or ensemble models.

Problem Description and Challenges Although FA and CE are both central to model interpretability and accountability, each is individually insufficient. FA in isolation assigns importance scores but

does not prescribe a concrete, executable set of feature modifications, while CE in isolation typically perturbs more features than are operationally practical. We therefore formulate a general problem that subsumes both settings and builds on the foundations established in the literature (see Appendix A). Concretely, we address the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{a}$	$\textcircled{b}$	$\textcircled{c}$	$\textcircled{a}$	$\textcircled{b}$	$\textcircled{c}$	$\textcircled{a}$	$\textcircled{b}$	$\textcircled{c}$	$\textcircled{c}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform seeks to increase user registrations. The observed data record user interaction covariates, namely the amount of money spent ($\textcircled{a}$), the number of clicks ($\textcircled{b}$), and a binary registration indicator ($\textcircled{c}$). In the factual data ($\mathbf{x}$), the registration indicator is zero for all users. The two action plans $\mathbf{z}'$ and $\mathbf{z}''$ perturb the user covariates so as to attain the target outcome ($\mathbf{y}^*$) of universal registration. The two plans induce the same (half) counterfactual effect, yet $\mathbf{z}''$ realizes it with strictly fewer coordinate modifications than $\mathbf{z}'$.

Addressing this problem entails three principal challenges. First, no single CE algorithm can satisfy all requirements simultaneously, because the objective is inherently task-specific. Second, to guarantee applicability across a broad class of predictors, the method must not impose strong structural assumptions on the model, such as differentiability or a prescribed architecture. Third, generic FA, such as marginal feature-importance scores, can be misleading, because the attribution scores are not coherent with the perturbations that actually induce the counterfactual effect. Stated precisely, computing FA independently of the CE and then selecting the top-scoring features to modify is ineffective. We show subsequently, in Result II of Section 6, that this decoupling yields counterproductive feature modifications (equivalently, the actions prescribed to users or stakeholders), since the features ranked as globally important need not lie on the specific pathways that realize the desired counterfactual outcomes.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that refines a given CE, for arbitrary predictive models and CE generators, by employing an optimal transport (OT)-induced coupling between the factual and counterfactual sets to condition Shapley attributions. The resulting attribution operator, p -SHAP, subsumes the commonly used Shapley instantiations as special cases obtained under specific choices of coupling, and exposes a modular interface separating attribution from edit selection. We provide two theoretical guarantees: the OT coupling minimizes an upper bound on the W_1 divergence, and, under a Lipschitz-continuity assumption, the refined counterfactuals are provably no farther from the factuals than the input CE. Empirically, across four datasets, twelve predictive models, and five CE methods, COLA equipped with p -SHAP attains identical target effects while retaining only 26–45% of the feature edits in the original CE. On a tractable small-scale benchmark, COLA closely matches the exact optimum.

To the best of our knowledge, this is the first method that systematically addresses the problem depicted in Figure 1 *without committing to a specific CE algorithm or a specific model class*.

2 PROBLEM FORMULATION

We now give a formal statement of the problem depicted in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model. Let $\mathbf{x}$ denote the observed (factual) data, arranged as n instances (rows) over d features (columns) with $\mathbf{x} \in \mathbb{R}^{n \times d}$ ($n \geq 1, d \geq 1$), and let $\mathbf{y}^*$ denote the target model output with $\mathbf{y}^* \in \mathbb{R}^m$ ($m \geq 1$). The task is to determine a set of counterfactual instances $\mathbf{z} \in \mathbb{R}^{n \times d}$, with each output $f(\mathbf{z})$ driven as close as possible to $\mathbf{y}^*$, subject to a budget of at most C feature modifications. Let D denote a divergence functional quantifying the dissimilarity between two arguments. The

problem is stated as follows.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) together with the constraint equation (1b) constitutes the canonical CE optimization: $\mathbf{z}$ is required to drive $f(\mathbf{z})$ toward $\mathbf{y}^*$ while remaining proximal to $\mathbf{x}$, so that $\mathbf{z}$ serves as a contrastive reference explaining why $f(\mathbf{x})$ fails to attain $\mathbf{y}^*$. We deliberately leave D unspecified so that the formulation remains general; admissible instantiations include the Euclidean distance, OT, maximum mean discrepancy (MMD), and the absolute difference in the mean or median of the model outputs. The constraints equation (1c)–equation (1e) augment the CE optimization with an explicit action budget. In addition to the counterfactual matrix $\mathbf{z}$, we introduce a binary modification-indicator matrix $\mathbf{c}$: coordinate z_{ik} is held fixed if and only if $c_{ik} = 0$, and equation (1c) caps the total number of modifications at C . The big- M constraints equation (1d) and equation (1e) encode this behavior: when $c_{ik} = 0$ they force $z_{ik} = x_{ik}$, so coordinate (i, k) is unaltered, whereas when $c_{ik} = 1$ the constant M is chosen sufficiently large that z_{ik} may vary freely.

To solve equation (1), we use FA to identify the most influential coordinates and thereby determine the modification-indicator matrix $\mathbf{c}$. The next section reviews the commonly used Shapley-value methods for FA which, together with the method we propose, are integrated into our algorithmic framework COLA to produce the refined counterfactual $\mathbf{z}$. The problem is computationally intractable: it remains hard even in the univariate case $d = 1$ with linear models (see Appendix B).

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews the commonly used Shapley-value methods for FA that, together with the operator we propose, are integrated into our algorithmic framework COLA. We first state the game-theoretic definition of the Shapley value and then specify the standard instantiations used for FA; we refer the reader to (Sundararajan & Najmi, 2020) for a comprehensive survey of these variants.

In cooperative game theory, a coalitional game is specified by a finite player set (here, the set of features) $\mathcal{F} = \{1, 2, \dots, d\}$ together with a characteristic function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ that assigns to each coalition $S \subseteq \mathcal{F}$ a real-valued payoff $v(S)$ attainable by the members of S through cooperation, normalized so that $v(\emptyset) = 0$. The Shapley value, defined below in terms of v , is the unique payoff-allocation rule that distributes the grand-coalition payoff $v(\mathcal{F})$ among the players subject to a standard set of fairness axioms.

Prior to defining the Shapley value, let $\Delta(k, S)$ denote the marginal contribution of player (feature) k to coalition S , i.e., the incremental payoff attributable to adjoining k to S ,

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i assigned to player i is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k is the average of the marginal contributions $\Delta(k, S)$ of feature k over all coalitions S not containing k , each weighted by the inverse binomial coefficient $\binom{d-1}{|S|}^{-1}$, where d is the total number of features. Summing over all such coalitions and dividing by d yields the axiomatic allocation of the payoff across features, so that ϕ_k quantifies the contribution of feature k relative to the full feature set.

For an arbitrary data point $\mathbf{x}_i$, we write $\mathbf{x}_{i,S}$ for its restriction to the features indexed by S and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ for its complementary restriction; the analogous convention applies to the remaining notation. We now specify the commonly used Shapley-value methods.

Baseline Shapley (B-SHAP). This instantiation fixes a single baseline point $\mathbf{r}_j$ as the reference for every $\mathbf{x}_i$, and defines the characteristic function as

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j, \mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

so that the absence of a feature is represented by substituting its value from the reference baseline $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This construction presupposes an exact one-to-one alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This variant of B-SHAP replaces the fixed reference by an expectation over a background distribution, and is the instantiation implicitly used in (Lundberg & Lee, 2017) and in the widely adopted ‘‘SHAP’’ library. Its characteristic function is

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} \left[f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S}) \right] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ denotes the background distribution, typically taken to be the empirical training distribution (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This method extends RB-SHAP by setting the background distribution $\mathcal{D}$ to the counterfactual distribution conditioned on each individual factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$),

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

It has been adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and shown advantageous for contrastive FA. This construction presupposes a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We unify equation (4) through equation (6) by parameterizing the characteristic function with an algorithm A_{Prob} that returns the joint probability of $\mathbf{x}$ and $\mathbf{r}$.¹ The p -SHAP characteristic function is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

The key distinction from prior instantiations, which condition on random baselines, is that p -SHAP determines the alignment explicitly by computing the optimal coupling via OT. This recasts feature attribution not as a mere decomposition of the prediction, but as a transport problem that minimizes the cost of the explanation.

We emphasize that A_{Prob} *does not depend on the CE algorithm and is a function only of the generated counterfactual $\mathbf{r}$* . By conditioning solely on these fixed quantities, p -SHAP yields FA that is invariant to the variability of the CE generation process—a property that distinguishes it from CF-SHAP. Contrary to what might be expected, we show that the OT coupling can be more effective than conditioning on a counterfactual distribution induced by a specific CE generation mechanism, as in Albini et al. (2022); see Result II of Section 6.

A central object in this paper is the entropically regularized OT problem (equivalently, the squared 2-Wasserstein divergence) defined below, whose optimal transport plan $\mathbf{p}_{\text{OT}}$ we adopt as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ within p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\mu, \nu)} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

¹The three standard instantiations are recovered as special cases. First, p -SHAP reduces to B-SHAP in equation (4) when A_{Prob} specifies a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that encodes a deterministic $i \leftrightarrow j$ alignment for every pair $\mathbf{x}_i, \mathbf{r}_j$. Second, it reduces to RB-SHAP in equation (5) when A_{Prob} is independent of the CE and instead associates $\mathbf{x}$ with an arbitrary distribution $\mathcal{D}$. Third, it reduces to CF-SHAP in equation (6) when A_{Prob} is built from a known counterfactual distribution.

Here μ and ν denote the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, respectively, and $\Pi(\mu, \nu)$ denotes the transport polytope, i.e., the set of joint distributions with these marginals. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularizer with coefficient $\varepsilon \geq 0$; a strictly positive coefficient ($\varepsilon > 0$) renders the problem strongly convex and admits efficient computation via the Sinkhorn–Knopp iteration.

Theoretical Aspects of p -SHAP Under the alignment it induces between the factual $\mathbf{x}$ and the counterfactual $\mathbf{r}$, the OT coupling minimizes the total feature-modification cost of transporting $\mathbf{x}$ toward $\mathbf{r}$, which is exactly the objective of achieving the counterfactual outcome at minimal cost. We formalize this connection under a Lipschitz-continuity assumption on the predictor f . Theorem 4.1 below (proof in Appendix C) establishes that the transport plan $\mathbf{p}_{OT}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, if f is L -Lipschitz, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is upper-bounded by L times the square root of the minimized expected transport cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where the weights p_{ij} ($j = 1, 2, \dots, m$) specify how the feature values of $\mathbf{x}_i$ are to be displaced toward those of one or several $\mathbf{r}_j$. Operationally, within p -SHAP the OT plan $\mathbf{p}_{OT}$ prescribes a displacement of the feature values of $\mathbf{x}$ toward those of $\mathbf{r}$ that minimizes the expected modification cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$. Among all admissible couplings $\mathbf{p} \in \Pi$, $\mathbf{p}_{OT}$ attains the minimal cost and hence furnishes the tightest such upper bound on the counterfactual-effect violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Theorem 4.1 bounds the transport cost, which acts as a convex surrogate for the $\mathcal{NP}$ -hard discrete sparsity (L_0) objective. By concentrating attribution mass on the lowest-cost transport paths, the OT coupling steers the greedy selection in Algorithm 1 toward discarding non-essential features. Moreover, p -SHAP admits a causal interpretation: it attributes the change in model output to interventions on the corresponding feature values, as formalized in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Finally, p -SHAP inherits the standard Shapley axioms satisfied by B-SHAP and RB-SHAP, which qualifies it as a principled attribution operator. We omit the proofs and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the statements of these axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The framework COLA, specified in Algorithm 1, targets problem equation (1) and is composed of four interchangeable algorithmic modules. We describe Algorithm 1 in detail below and illustrate it in Figure 2.

Line 1 (*Generating a counterfactual $\mathbf{r}$ via a CE algorithm*). The CE algorithm A_{CE} receives the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ , and returns a counterfactual $\mathbf{r}$ proximal to $\mathbf{x}$ satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

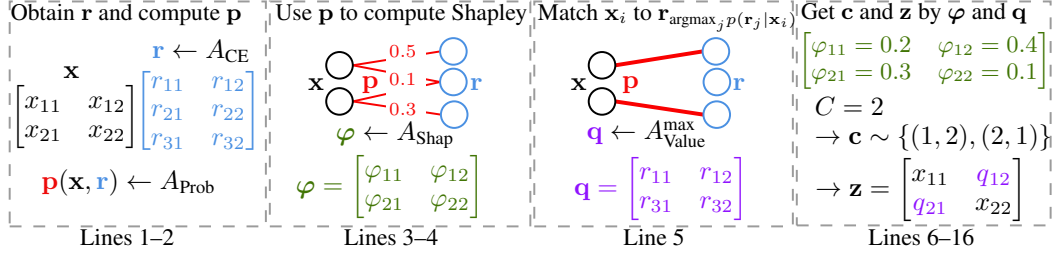


Figure 2: [An illustration of COLA] This figure shows how COLA gets $\mathbf{c}$ and $\mathbf{z}$ for equation (1). We use $A_{\text{Value}}^{\max}$ for illustration in line 5 due to its simplicity. In lines 6–16, we assume $C = 2$, and the sampling yields exactly two positions for modifications according to the probability matrix φ .

Line 2 (*Estimating the joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). The module A_{Prob} receives $\mathbf{x}$ and $\mathbf{r}$ and returns a matrix encoding the joint distribution over the n rows of the factual $\mathbf{x}$ and the m rows of the counterfactual $\mathbf{r}$. This joint distribution $\mathbf{p}$ specifies an alignment (matching) between the factual rows and the counterfactual rows, and is consumed in Line 5 to compute the reference values from which $\mathbf{z}$ is subsequently assembled. As discussed in Section 3, when the row alignment is unknown, A_{Prob} can be instantiated via OT to yield the coupling attaining the minimal OT distance between $\mathbf{x}$ and $\mathbf{r}$; when a reliable alignment is available, A_{Prob} should instead return a coupling that reflects it.

Lines 3–4 (*p-SHAP FA*). We evaluate equation (7) to obtain the Shapley values for each row of the factual data $\mathbf{x}$, thereby attributing the model output over $\mathbf{x}$. The joint distribution $\mathbf{p}$ may (optionally) be used here to align each row of the factual $\mathbf{x}$ with its corresponding counterfactual rows of $\mathbf{r}$, so that the selected rows of $\mathbf{r}$ constitute the most representative contrastive reference for that row of $\mathbf{x}$. By coupling the rows of $\mathbf{x}$ with those of $\mathbf{r}$ in this manner, the alignment becomes, empirically, a decisive factor in the quality of the contrastive FA. The resulting Shapley-value matrix is then transformed element-wise by taking absolute values and normalizing to sum to one, yielding the attribution matrix φ .

Line 5 (*Computing reference feature values*). The module A_{Value} receives the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$. For each row i of the factual $\mathbf{x}$, it selects one or several rows of the counterfactual $\mathbf{r}$ to serve as references for modifying that row of $\mathbf{x}$, and returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ whose entry q_{ik} is the candidate counterfactual value for x_{ik} ($\forall i, k$). We specify two instantiations, $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$, corresponding respectively to single-row and multi-row selection.

For each row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ selects the row of $\mathbf{r}$ assigned the highest coupling probability,

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The instantiation A^{avg} computes q_{ik} as a convex combination (coupling-weighted average) of $r_{1k}, r_{2k}, \dots, r_{mk}$,

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Since the non-negative attribution matrix φ is normalized to sum to one, it defines a sampling distribution over coordinates, which we use as the selection policy for value replacement (i.e., setting $c_{ik} = 1$) in line 7. For every (i, k) with $c_{ik} = 1$, the entry x_{ik} is reassigned to q_{ik} ; the resulting matrix is returned as $\mathbf{z}$, and together with $\mathbf{c}$ it constitutes the solution to problem equation (1).

By Theorem 4.1, COLA minimizes the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by modifying $\mathbf{z}$ at the coordinates identified by the feature attribution as most influential for realizing the desired counterfactual effect. The theorem below (proof in Appendix E) establishes that the refined $\mathbf{z}$ produced by COLA satisfies the proximity constraint equation (1b) in the canonical case $n = m$ under the Frobenius norm. Empirical evidence that the conclusion holds more generally is reported in Table 3.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ denote the aligned reference matrix obtained from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ through the value-assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, with $\mathbf{p}$ the coupling matrix. Then, for all dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ produced by COLA satisfies*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

That is, $\mathbf{z}$ is no farther from the factual input $\mathbf{x}$ than the aligned counterfactual proposal $\mathbf{q}$, so the refinement step introduces no additional divergence.

Corollary 5.2. *Consider the special case $n = m$ in which the OT plan $\mathbf{p}$ is a deterministic permutation σ (obtained without entropic regularization, $\varepsilon = 0$). Then $\mathbf{q}$ coincides with the row-permutation of $\mathbf{r}$, denoted $\mathbf{r}_\sigma$ with $q_i = r_{\sigma(i)}$, and the bound of Theorem 5.1 specializes to*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

i.e., the refined counterfactual is no farther from the facts than the original counterfactual set reordered under σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the complexity of A_{CE} . For A_{Shap} , we adopt the weighted-linear-regression estimator of Shapley values and let M_{Shap} denote the number of sampled coalitions. The overall complexity of COLA, as a function of n, m, d , and the entropic-OT regularization parameter ε , is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ under $A_{\text{Value}}^{\max}$ and $N = \mathcal{O}(nmd)$ under $A_{\text{Value}}^{\text{avg}}$. A full derivation is given in Appendix F.

6 NUMERICAL RESULTS

This section empirically evaluates COLA on problem equation (1), with the target set to $\mathbf{y}^* = f(\mathbf{r})$, where $\mathbf{r}$ is the counterfactual returned by a CE method A_{CE} . We instantiate the divergence functional D with four choices: OT measures the distance between the full distributions; MMD measures the discrepancy between the distribution means in a reproducing-kernel feature space; and absolute mean difference (MeanD) and absolute median difference (MedianD) measure the absolute difference in the mean and median of the outputs, respectively. The experiments are designed to establish three claims: I) COLA is effective at minimizing the number of modifications. II) p -SHAP attains superior modification minimality relative to competing Shapley methods. III) COLA closely approaches the exact optimum.

Experiment Setup The evaluation spans 4 binary-classification datasets, 5 CE algorithms targeting distinct objectives, and 12 classifiers, enumerated in Table 1. We define an “experiment scenario” as a triple consisting of a dataset, a CE algorithm A_{CE} , and a model f .

Table 1: [Experiment Scenarios Setup] Four datasets, five CE algorithms, and 12 models are used for benchmarking. A “scenario” denotes a triple comprising a dataset, a CE algorithm A_{CE} , and a model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] Specification of the 6 compared methods, color-coded consistently with Figure 3. Each method is evaluated within the experiment scenarios defined in Table 1.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

We summarize the CE algorithms A_{CE} listed in Table 1. DiCE and KNN are instance-based methods that return one or more counterfactuals per factual instance. AReS and GlobeCE are group-based methods that return a collection of counterfactuals for the factual set treated as a whole. Discount models the factual instances as an empirical distribution and seeks a counterfactual distribution proximal to it.

Table 2 specifies the 6 compared methods, of which CF- p_{OT} is the proposed p -SHAP and the remaining five are baselines; each is benchmarked across many scenarios from Table 1. A method is fully determined by its choice of the three modules A_{Prob} , A_{Value} , and A_{Shap} . For instance, RB- p_{Uni} (first row) instantiates A_{Prob} with the uniform coupling $\mathbf{p}$, propagates this $\mathbf{p}$ into A_{Value} via equation (9) or equation (10) to form $\mathbf{q}$, and sets A_{Shap} to RB-SHAP.

These methods are constructed to isolate individual factors in a controlled ablation. First, RB- p_{Uni} and CF- p_{Uni} differ only in that the latter incorporates CE information whereas the former does not. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all incorporate the CE into FA but under different couplings, allowing us to show that the OT coupling significantly outperforms the alternatives. Third, to verify that the benefit of OT stems specifically from the factual-counterfactual alignment it supplies to A_{Shap} rather than from confounding factors, we compare CF- p_{OT} , which uses this alignment, against RB- p_{OT} , which does not (the two differ only in A_{Shap}). Finally, CF- p_{Ect} is the idealized case in which each counterfactual has a known source, providing an exact factual-counterfactual alignment under which CF-SHAP coincides with B-SHAP.

Result I: COLA attains substantial action reduction at negligible loss of counterfactual effect

In Table 3, we constrain $\mathbf{z}$ so that $f(\mathbf{z})$ attains 80% or 100% of the counterfactual effect of $f(\mathbf{r})^2$. COLA attains this target while requiring markedly fewer actions in $\mathbf{z}$ (i.e., feature modifications within $\mathbf{z}$) than the original CE $\mathbf{r}$: reaching the 80% counterfactual effect requires only 13%–25% of the feature changes, as measured by the ratio $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$. Furthermore, COLA with p -SHAP (i.e., CF- p_{OT}) is the only method that attains the full 100% counterfactual effect, and does so using

²By construction $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which corresponds to a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. We define the counterfactual effect as the fraction of the divergence eliminated by the refined CE $\mathbf{z}$, i.e., $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*)$, with D set to OT; the 80% target corresponds to this quantity equalling 0.80.

Table 3: [COLA for Modification Minimality] Number of modified features in $\mathbf{z}$ per method at the point where $f(\mathbf{z})$ attains 80% (and 100%) of the counterfactual effect of $f(\mathbf{r})$. Each entry is averaged over 4 scenarios drawn at random from Table 1, with 100 runs per scenario. The symbol “-” indicates that the target counterfactual effect is unattainable.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

only 26%–45% of the feature changes present in $\mathbf{r}$. Thus p -SHAP yields the strongest modification minimality; we analyze this in detail below.

Result II: p -SHAP outperforms competing Shapley methods in realizing the counterfactual effect Figure 3 reports the Shapley instantiations of equation (4)–equation (7), plotting the objective $D(f(\mathbf{z}), \mathbf{y}^*)$ of equation (1) (ordinate) against the action budget C (abscissa). Three observations follow. First, RB- p_{Uni} and RB- p_{OT} perform substantially worse than the remaining methods, confirming that incorporating CE information into the attribution is essential for modification minimality. Second, since CF- p_{OT} and RB- p_{OT} differ only in A_{Shap} (see Table 2), the fact that CF- p_{OT} dominates RB- p_{OT} isolates the benefit of OT to the factual-counterfactual alignment it supplies, rather than to any confounding factor. Third, p -SHAP markedly outperforms CF- p_{Uni} and CF- p_{Rnd} , establishing the value of the OT coupling: conditioning on counterfactual information alone (as in CF- p_{Uni} and CF- p_{Rnd}) is insufficient, and an appropriate factual-counterfactual alignment must be used—one that, as Figure 4 shows, need not coincide with the exact counterfactual-generation mechanism.

As shown in Figure 3 and Table 3, COLA with p -SHAP performs strongly across the scenarios of Table 1; an extensive set of additional results appears in Appendix H. We conclude that p -SHAP dominates all competing Shapley methods with respect to sparsity in CE.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks This result shows that p -SHAP eliminates the dependence on the CE generation process by substituting the OT joint distribution between factual and counterfactual data for the CE-algorithm-dependent knowledge³ of $\mathcal{D}$; see Figure 4. We benchmark CF- p_{Ect} under COLA, solving equation (1) with D set to MeanD. Because CF- p_{Ect} presupposes a known factual-counterfactual alignment, this experiment quantifies how effectively COLA exploits such an alignment. The exact optimum in this setting is obtained by solving an MILP (Appendix G derives the formulation); as solving the MILP is computationally expensive, we restrict this comparison to *German Credit*.

³This knowledge—an exact factual-counterfactual alignment—is available only for DiCE and KNN among the CE methods considered.

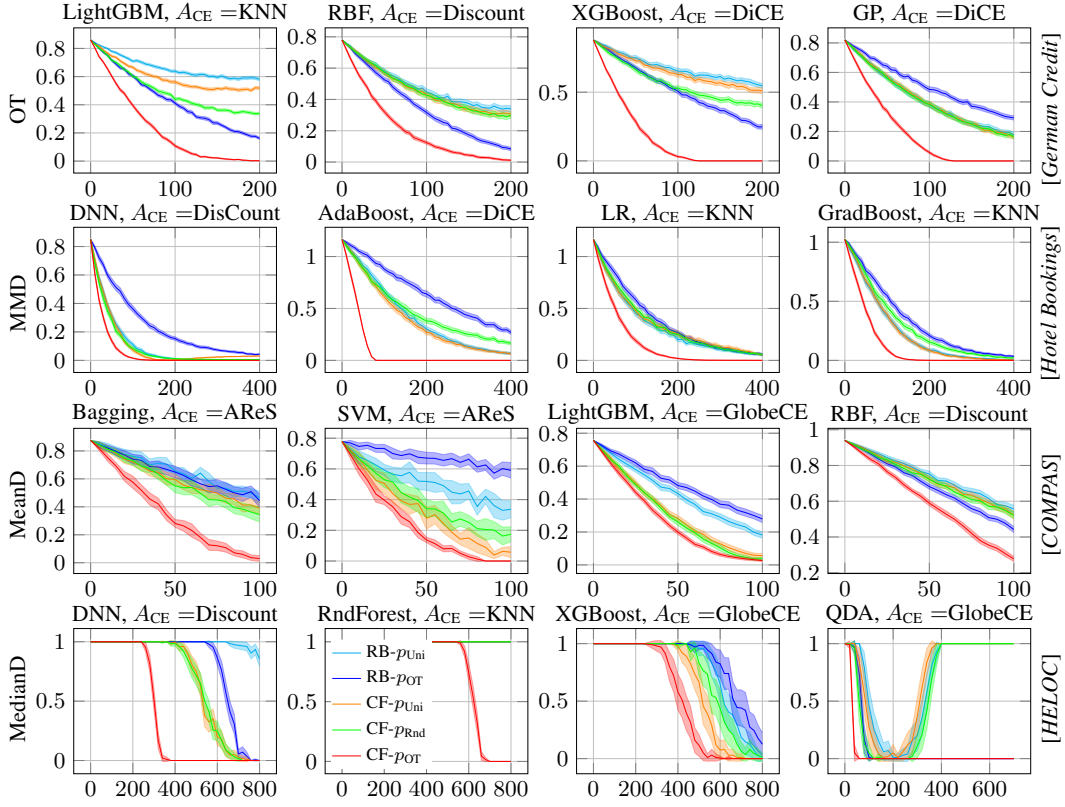


Figure 3: Divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C . Each curve aggregates 100 runs, and the shaded band denotes the 99.9% confidence interval. The value module is $A_{\text{Value}}^{\text{avg}}$ for *HELOC* and *COMPAS* and $A_{\text{Value}}^{\text{max}}$ for *German Credit* and *Hotel Bookings*. The legend shown in the “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” panel applies to all panels.

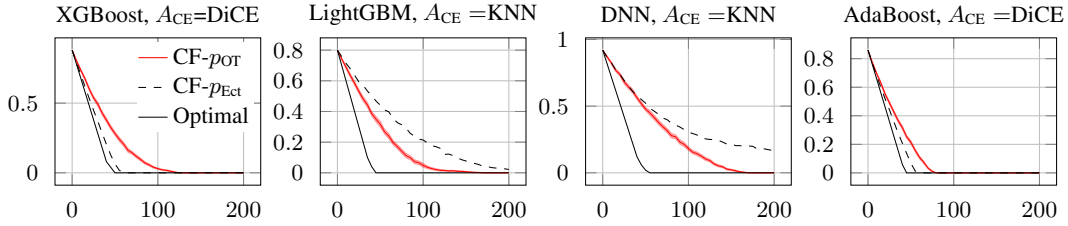


Figure 4: [*German Credit*] Divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C , with D instantiated as *MeanD*.

CF- p_{Ect} attains near-optimal performance under DiCE. Since the factual-counterfactual pairs are mutually independent for DiCE and KNN, we conclude that COLA is effective for instance-based CE, even though formulation equation (1) is stated for the more general group or distributional setting. Notably, CF- p_{OT} occasionally outperforms CF- p_{Ect} , because it identifies the features requiring modification through a more theoretically grounded criterion, whereas CF- p_{Ect} relies on CE-algorithm-dependent knowledge whose effectiveness for FA lacks a rigorous justification. A residual gap between CF- p_{Ect} , CF- p_{OT} , and the exact optimum remains under KNN; determining the best alignment is left as an open problem.

7 CONCLUSIONS

This paper introduces COLA, a framework that refines CE via joint-distribution-informed Shapley values, so that the refined CE preserves the counterfactual effect while requiring fewer actions. The framework is agnostic to the predictive model and the CE generator, is supported by theoretical guarantees on both the W_1 bound and the proximity of the refined counterfactual, and is validated empirically across four datasets, twelve models, and five CE generators.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a method (COLA with p -SHAP) for refining CE. Our work is methodological and empirical; it does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We follow dataset licenses/terms and the common research practice of using these data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility/immutability constraints on features.

CE can surface biases in underlying models but do not by themselves guarantee fairness. Our method can reduce the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age) nor does it confer causal validity. We discourage use of CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should include domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact/benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional (not causal) and must not be interpreted as causal effect.

Risks include (i) misinterpretation of attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these by (a) stating non-causal scope and limitations in the paper, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing guidance for responsible use in documentation.

We provide an anonymized implementation, detailed settings, and seeds to facilitate replication; we report model types, hyperparameters, and compute budgets, and we avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We provide pointers to all components required for independent replication. The COLA framework and the p -SHAP attribution operator are fully specified via pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), with all theoretical assumptions stated explicitly and complete, step-by-step proofs deferred to the appendix.

The implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which also specifies the datasets and the full evaluation protocol, including data splits and preprocessing. The baselines (predictive models and CE generators) and the coupling choices are enumerated in the experimental sections and ablations, and their configurations are mirrored in the supplementary materials. The experiment code and software will be released upon acceptance; Appendix I additionally reports the hardware and runtime to contextualize the compute requirements, and the appendix contains the ablation and sensitivity studies together with the small-scale MILP benchmark setup to support independent verification.

⁴<https://iclr.cc/public/CodeOfEthics>

REFERENCES

- Google OR-Tools. URL <https://developers.google.com/optimization>.
- Gurobi Optimization, LLC. URL <https://www.gurobi.com>.
- Machine Learning in Python. URL <https://scikit-learn.org/stable/>.
- Emanuele Albini, Jason Long, Danial Dervovic, and Daniele Magazzeni. Counterfactual shapley additive explanations. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pp. 1054–1070, 2022.
- Nuno Antonio, Ana de Almeida, and Luis Nunes. Hotel booking demand datasets. *Data in brief*, 22: 41–49, 2019.
- Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Bennetot, Siham Tabik, Alberto Barbado, Salvador García, Sergio Gil-López, Daniel Molina, Richard Benjamins, et al. Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai. *Information fusion*, 58:82–115, 2020.
- Emilio Carrizosa, Jasone Ramírez-Ayerbe, and Dolores Romero Morales. Mathematical optimization modelling for group counterfactual explanations. *European Journal of Operational Research*, 2024.
- Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794, 2016.
- Claudio Contardo, Ricardo Fukasawa, Louis-Martin Rousseau, and Thibaut Vidal. Optimal counterfactual explanations for k-nearest neighbors using mathematical optimization and constraint programming. URL <https://optimization-online.org/>.
- Arun Das and Paul Rad. Opportunities and challenges in explainable artificial intelligence (xai): A survey. *arXiv preprint arXiv:2006.11371*, 2020.
- FICO. Fico explainable machine learning challenge, 2018. URL <https://community.fico.com/s/explainable-machine-learning-challenge>.
- Alexandre Forel, Axel Parmentier, and Thibaut Vidal. Explainable data-driven optimization: From context to decision and back again. In *International Conference on Machine Learning*, pp. 10170–10187. PMLR, 2023.
- Riccardo Guidotti. Counterfactual explanations and how to find them: literature review and benchmarking. *Data Mining and Knowledge Discovery*, pp. 1–55, 2022.
- Hans Hofmann. Statlog (German Credit Data). UCI Machine Learning Repository, 1994. DOI: <https://doi.org/10.24432/C5NC77>.
- Lauren Kirchner Jeff Larson, Surya Mattu, and J Angwin. How we analyzed the compas recidivism algorithm, 2016. URL <https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm>.
- Kentaro Kanamori, Takuya Takagi, Ken Kobayashi, and Hiroki Arimura. Dace: Distribution-aware counterfactual explanation by mixed-integer linear optimization. In *IJCAI*, pp. 2855–2862, 2020.
- Kentaro Kanamori, Takuya Takagi, Ken Kobayashi, and Yuichi Ike. Counterfactual explanation trees: Transparent and consistent actionable recourse with decision trees. In *International Conference on Artificial Intelligence and Statistics*, pp. 1846–1870. PMLR, 2022.
- Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from counterfactual explanations to interventions. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, pp. 353–362, 2021.

648 Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan
649 Liu. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information*
650 *processing systems*, 30, 2017.

651
652 Ramaravind Kommiya Mothilal, Divyat Mahajan, Chenhao Tan, and Amit Sharma. Towards uni-
653 fying feature attribution and counterfactual explanations: Different means to the same end. In
654 *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pp. 652–663, 2021.

655 Yongchan Kwon and James Y Zou. Weightedshap: analyzing and improving shapley based feature
656 attributions. *Advances in Neural Information Processing Systems*, 35:34363–34376, 2022.

657
658 Dan Ley, Saumitra Mishra, and Daniele Magazzeni. Global counterfactual explanations: Investiga-
659 tions, implementations and improvements. *arXiv preprint arXiv:2204.06917*, 2022.

660 Dan Ley, Saumitra Mishra, and Daniele Magazzeni. Globe-ce: a translation based approach for global
661 counterfactual explanations. In *International Conference on Machine Learning*, pp. 19315–19342.
662 PMLR, 2023.

663
664 Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in*
665 *neural information processing systems*, 30, 2017.

666
667 Luke Merrick and Ankur Taly. The explanation game: Explaining machine learning models using
668 shapley values. In *Machine Learning and Knowledge Extraction: 4th IFIP TC 5, TC 12, WG*
669 *8.4, WG 8.9, WG 12.9 International Cross-Domain Conference, CD-MAKE 2020, Dublin, Ireland,*
670 *August 25–28, 2020, Proceedings 4*, pp. 17–38. Springer, 2020.

671
672 Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. Explaining machine learning classifiers
673 through diverse counterfactual explanations. In *Proceedings of the 2020 conference on fairness,*
674 *accountability, and transparency*, pp. 607–617, 2020.

675
676 Yasunobu Nohara, Koutarou Matsumoto, Hidehisa Soejima, and Naoki Nakashima. Explanation
677 of machine learning models using shapley additive explanation and application for real data in
678 hospital. *Computer Methods and Programs in Biomedicine*, 214:106584, 2022.

679
680 Axel Parmentier and Thibaut Vidal. Optimal counterfactual explanations in tree ensembles. In
681 *International conference on machine learning*, pp. 8422–8431. PMLR, 2021.

682
683 Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor
684 Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style,
685 high-performance deep learning library. *Advances in neural information processing systems*, 32,
686 2019.

687
688 Kaivalya Rawal and Himabindu Lakkaraju. Beyond individualized recourse: Interpretable and
689 interactive summaries of actionable recourses. *Advances in Neural Information Processing Systems*,
690 33:12187–12198, 2020.

691
692 Amit Sharma, Hua Li, and Jian Jiao. The counterfactual-shapley value: Attributing change in system
693 metrics. In *NeurIPS 2022 Workshop on Causality for Real-world Impact*.

694
695 Yi Sun and Mukund Sundararajan. Axiomatic attribution for multilinear functions. In *Proceedings of*
696 *the 12th ACM conference on Electronic commerce*, pp. 177–178, 2011.

697
698 Mukund Sundararajan and Amir Najmi. The many shapley values for model explanation. In
699 *International conference on machine learning*, pp. 9269–9278. PMLR, 2020.

700
701 Lucile Ter-Minassian, Sahra Ghalebikesabi, Karla Diaz-Ordaz, and Chris Holmes. Challenges and
opportunities of shapley values in a clinical context. *arXiv preprint arXiv:2306.14698*, 2023.

Berk Ustun, Alexander Spangher, and Yang Liu. Actionable recourse in linear classification. In
Proceedings of the conference on fairness, accountability, and transparency, pp. 10–19, 2019.

Sahil Verma, John Dickerson, and Keegan Hines. Counterfactual explanations for machine learning:
A review. *arXiv preprint arXiv:2010.10596*, 2, 2020.

702 Sandra Wachter, Brent Mittelstadt, and Chris Russell. Counterfactual explanations without opening
703 the black box: Automated decisions and the gdpr. *Harv. JL & Tech.*, 31:841, 2017.
704
705 Lei You, Lele Cao, Mattias Nilsson, Bo Zhao, and Lei Lei. Distributional counterfactual explanations
706 with optimal transport. In *The 28th International Conference on Artificial Intelligence and Statistics*,
707 2025.
708
709
710
711
712
713
714
715
716
717
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A COMPARISON TO EXISTING APPROACHES

Albini et al. (2022) proposed CF-SHAP, which employs counterfactual data points as the background distribution for the Shapley computation. This construction assumes that the counterfactual distribution is specified conditionally on each individual instance, which presupposes a known probabilistic alignment between every factual and every counterfactual instance; analogous assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). This assumption is violated in settings that demand global explanations. The counterfactual-distribution setup of (Albini et al., 2022) is a special case of ours. Kwon & Zou (2022) proposed WeightedSHAP, which assigns non-uniform weights across features rather than treating their contributions as equal; our method instead weights contributions across data points and extends straightforwardly to joint weighting over both rows and columns. Kommiya Mothilal et al. (2021) establish a framework that couples FA and CE for explainability, but their CE-based FA inherits the assumption of (Albini et al., 2022) and therefore does not generalize to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). A further related method, (Sharma et al.), models counterfactuals through a system’s causal structure and time-series predictors in order to attribute a single observed change in a system-level metric to the individual inputs. Critically, none of the foregoing works addresses the minimal-action CE problem that is the focus of this paper.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) are the closest to ours. Karimi et al. (2021) minimize the cost of the prescribed actions under the assumption of a known structural causal model (SCM), which is seldom available in practice. Kanamori et al. (2022) observe that whether existing CE methods can solve this problem remains open, and propose an MILP-solver-based approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Solving the MILP, however, is computationally expensive and consequently scales poorly.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below shows that, unless $\mathcal{P} = \mathcal{NP}$, no scalable exact algorithm for the general problem can be expected. The hardness is not attributable solely to the non-linearity of f ; it persists because of the combinatorial nature of the coordinate-selection constraint.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. We reduce from the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the objective is to find $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error subject to having at most K non-zero

entries:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero entries of $\mathbf{z}$. This problem is $\mathcal{NP}$ -hard owing to the combinatorial cost of selecting the subset of variables to retain in the model.

We construct a polynomial-time reduction from this SR problem to our problem equation (1) under the following instantiation. Set the number of features to $d = 1$ and the number of instances to n (matching the dimension of the SR problem). Take the factual data to be the zero vector $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$, and the target output to be $\mathbf{y}^* \in \mathbb{R}^m$ as given. Define the model as the linear map $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the output divergence, set $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$, the squared Euclidean distance; for the instance divergence, set $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, using $\mathbf{x} = \mathbf{0}$. Set the modification budget C equal to the sparsity level k of the SR problem. The big- M constant may be any sufficiently large positive number, e.g., $M \geq \max_i |z_i|$.

Since $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) reduce to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Hence $c_i = 0$ forces $z_i \leq 0$ and $z_i \geq 0$, i.e., $z_i = 0$, whereas $c_i = 1$ permits $z_i \in [-M, M]$. Constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Problem equation (1) thus reduces to:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation the binary variable c_i encodes whether z_i may change ($c_i = 1$) or is pinned to zero ($c_i = 0$); the constraints enforcing $z_i = 0$ whenever $c_i = 0$ reproduce the support constraint of the SR problem, and $\sum_{k=1}^n c_i \leq k$ caps the support size at the sparsity level k . The objective is identical to that of SR.

Because this construction coincides exactly with cardinality-constrained SR, which is $\mathcal{NP}$ -hard, solving problem equation (1) is at least as hard. Consequently, problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, f is linear, and both divergences D are squared Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ denotes the set of couplings satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The following theorem establishes that the feature attributions are aligned with the expected feature-modification costs, so that the induced action plans are cost-efficient in achieving the counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. By the L -Lipschitz continuity of f , for all $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$,

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides by $p_{ij}^{OT} \geq 0$ yields

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing over all $i = 1, \dots, n$ and $j = 1, \dots, m$ gives

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Define $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, so that the inequality reads

$$E_f \leq L E_d.$$

We now bound E_d via the Cauchy-Schwarz inequality. The weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$, since p_{OT} is a probability distribution on the joint space of $\mathbf{x}$ and $\mathbf{r}$. Recall that, for any real-valued arrays a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Choosing $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$ gives

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying Cauchy-Schwarz,

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this reduces to

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking square roots,

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the bound on E_f yields

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Thus the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by L times the square root of the expected transport cost under $\mathbf{p}_{\text{OT}}$. Finally, because $\mathbf{p}_{\text{OT}}$ minimizes the expected transport cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible couplings in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$,

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

This establishes the claim. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ denote the joint probability of $\mathbf{x}$ and $\mathbf{r}$ returned by A_{prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are clamped to $\mathbf{x}_{i,S}$, while those in $\mathcal{F} \setminus S$ remain distributed according to the marginal $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Hence the expected value of $f(\mathbf{r})$ under the intervention is

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

By the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

This establishes the claim. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ under the intervention that sets the features in S to $\mathbf{x}_{i,S}$, denoted $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This interventional quantity does not depend on any prescribed joint distribution $p(\mathbf{x}, \mathbf{r})$; accordingly, $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ decomposes into the base expectation of $f(\mathbf{r})$ plus the additional causal contribution of the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ denote the aligned reference matrix produced by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. At the modified coordinates ($c_{ik} = 1$) we have $z_{ik} = q_{ik}$, and at the unmodified coordinates ($c_{ik} = 0$) we have $z_{ik} = x_{ik}$. Consequently,

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms relative to $\mathbf{q}$ and $\mathbf{z}$ are

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where equality (i) follows from equation (12).

Since $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, we obtain the following inequality.

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots then yields

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This establishes the claim. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Proof. Under $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ is a permutation matrix induced by a bijection σ . The value-assignment module A_{value} , in either its maximizing or its averaging form, therefore returns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting the permuted matrix $\mathbf{r}_\sigma$ for $\mathbf{q}$ in the general bound of Theorem 5.1, namely $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$, gives the claim. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We analyze the computational complexity of COLA module by module.

First, consider A_{Prob} . When the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, A_{Prob} merely populates the matrix $\mathbf{p}$ from this prior, at cost $\mathcal{O}(n \times m)$. Otherwise, computing $\mathbf{p}$ by solving the entropic OT problem with the Sinkhorn–Knopp algorithm costs $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Next, consider A_{Shap} . Evaluating a single coalition requires evaluating the model on all n data points, i.e., $\mathcal{O}(n)$ evaluations per coalition. Substituting baseline values from the reference data $\mathbf{r}$ requires overwriting the corresponding feature values by accessing the baselines in the reference table $\mathbf{r}$; under the assumption that these reference values are precomputed and accessed in constant time, this substitution costs $\mathcal{O}(d)$. Aggregating over M_{Shap} sampled coalitions yields a total cost of $\mathcal{O}(n \times d \times M_{\text{Shap}})$ for A_{Shap} .

The normalization in line 4 of COLA costs $\mathcal{O}(n \times d)$.

We now compare $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$ step by step. For $A_{\text{Value}}^{\text{max}}$, each row $\mathbf{x}_i$ requires (1) computing p_{ij} for all $j \in \{1, \dots, m\}$, at $\mathcal{O}(m)$ per row; (2) identifying the reference row $\mathbf{r}_j$ of maximal probability p_{ij} , at $\mathcal{O}(m)$ per row; and (3) assigning $q_{ik} = r_{\tau(i),k}$ with $\tau(i) = \arg \max_j p_{ij}$, at $\mathcal{O}(d)$ per row. Over all n rows, the total cost is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, each row $\mathbf{x}_i$ requires (1) computing p_{ij} for all $j \in \{1, \dots, m\}$, at $\mathcal{O}(m)$ per row; (2) computing the normalizer $\sum_{j'=1}^m p_{ij'}$, at $\mathcal{O}(m)$ per row; and (3) forming the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, at $\mathcal{O}(m \times d)$ per row. Over all n rows, the total cost is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the cost is $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$, since $C \leq n \times d$.

Aggregating these terms, the complexity of COLA under $A_{\text{Value}}^{\text{max}}$ is

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA under $A_{\text{Value}}^{\text{avg}}$ is

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the overall complexity of COLA, as a function of n , m , d , and the entropic-OT regularization parameter ε , is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ under $A_{\text{Value}}^{\text{max}}$ and $N = \mathcal{O}(nmd)$ under $A_{\text{Value}}^{\text{avg}}$.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

In this section, we derive a global-optimality benchmark for solving equation (1) with a known factual-counterfactual alignment and D set to MeanD, i.e.,

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

where $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Because COLA guarantees $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$ together with $\mathbf{z}$ being no farther from $\mathbf{x}$ than $\mathbf{r}$, the proximity constraint equation (1b) is automatically satisfied and can be dropped.

Problem equation (1) then specializes to

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

The original big- M constraints equation (1d) and equation (1e) collapse into the single constraint equation (13c) because CF- p_{Ect} is used: for each $\mathbf{x}_i$ there is an exact reference $\mathbf{r}_j$ within A_{Shap} and A_{Value} , so each x_{ik} ($k = 1, \dots, d$) is either retained or set to r_{jk} . Hence $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$, whose value is determined by the binary variable c_{ik} .

Because the alignment between each $\mathbf{x}_i$ and its reference $\mathbf{r}_j$ is known, the reference matrix $\mathbf{q}$ is uniquely determined, and $A_{\text{Values}}^{\text{max}}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$. For each instance i and each feature subset $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denote the candidate solution $\mathbf{z}_i$ in which every feature $k \in S$ is set to q_{ik} and every feature $h \in \mathcal{F} \setminus S$ retains x_{ih} . The collection $\{\mathbf{z}_{iS} : S \subseteq \mathcal{F}\}$ therefore enumerates the domain of admissible values of $\mathbf{z}$. For each $\mathbf{z}_{iS}$, define the scalar

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be expressed through binary selection variables a_{iS} and the scalars g_{iS} as

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

Problem equation (13) is thereby reformulated as the following mixed-integer linear program.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η subject to equation (14b) and equation (14c) is equivalent to minimizing the absolute-value objective of equation (13). Constraint equation (14d) enforces that each instance i is assigned exactly one modification plan a_{iS} for a single subset $S \subseteq \mathcal{F}$, and constraint equation (14e) is the action-budget counterpart of equation (13b). Solving equation (14) yields the exact optimum of COLA under a known factual-counterfactual alignment, as reported in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

Across Figures 5–8, CF- p_{Uni} generally outperforms RB- p_{Uni} . Similarly, RB- p_{OT} and CF- p_{OT} differ only in A_{Shap} , and the latter consistently dominates the former. We therefore conclude that *RB-SHAP is unsuitable for FA in CE*.

We analyze the effect of the Shapley method on FA, corresponding to lines 3–4 of COLA, via two controlled comparisons. First, RB- p_{Uni} and CF- p_{Uni} differ only in A_{Shap} ; across Figures 5–8, CF- p_{Uni} generally outperforms RB- p_{Uni} . Second, RB- p_{OT} and CF- p_{OT} likewise differ only in A_{Shap} , and the latter consistently dominates. Both comparisons confirm that *RB-SHAP is unsuitable for FA in CE*.

Beyond FA, the second critical step of COLA is line 5, which uses the joint probability $p(\mathbf{x}, \mathbf{r})$ to construct the reference matrix $\mathbf{q}$, thereby specifying the displacement direction along which the factual x is moved toward the target model output. Figures 5–8 show that CF- p_{OT} consistently outperforms all competing methods across all experiments. All three of CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} furnish a coupling $\mathbf{p}$ in the absence of a known exact factual-counterfactual alignment, yet their performance differs markedly, which shows that knowledge of the CE and its marginal distribution alone is insufficient.

OT is especially valuable when the alignment between factual and counterfactual instances is unavailable or unreliable. Even when a CE algorithm explicitly matches each factual instance to a counterfactual, there is no guarantee that this known alignment is performance-optimal, as evidenced by Figure 4 in Section 6.

We stress that p_{OT} need not be the generative joint distribution of $\mathbf{x}$ and $\mathbf{r}$; its role is to guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ jointly in both FA and CE. Moreover, the QDA column of Figure 5 illustrates the stability of the OT-based methods, in contrast to the other methods, which diverge substantially from the target. Nonetheless, COLA is *not restricted to OT* as its choice of A_{Prob} . As Figure 4 indicates, even the best known $\mathbf{p}$ retains a non-negligible gap to the global optimum; designing an improved A_{Prob} is thus a promising direction for future work.

I EXPERIMENTS REPRODUCIBILITY

The experiments are executed on a high-performance-computing (HPC) cluster using four nodes in parallel, one per dataset, each node equipped with two Intel Xeon 2660v3 processors (10 cores, 2.60 GHz) and 128 GB of memory. Per-node wall-clock time is approximately 5–10 hours, depending on dataset size. The experiments are also reproducible on a single laptop, albeit at generally higher wall-clock cost than on the HPC cluster.

For all four datasets, numerical features are standardized and categorical features are encoded by either label encoding or one-hot encoding; we observed no appreciable difference in COLA’s performance between the two encodings. The train-test split ratio is 7:3.

The optimality baseline in Figure 4 is computed with Gurobi 11.0.2 ([gur](#)), which requires a valid license; alternatively, open-source operations-research solvers such as Google OR-Tools ([goo](#)) may be used. As noted, solving the MILP of Appendix G is computationally expensive and is therefore practical only for small-scale datasets such as German Credit; computing the baseline for larger datasets requires reducing the number of features.

The model hyperparameters are as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn implementations ([skl](#)) with default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN comprises three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are accessed through their scikit-learn interfaces with default hyperparameters.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assist tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

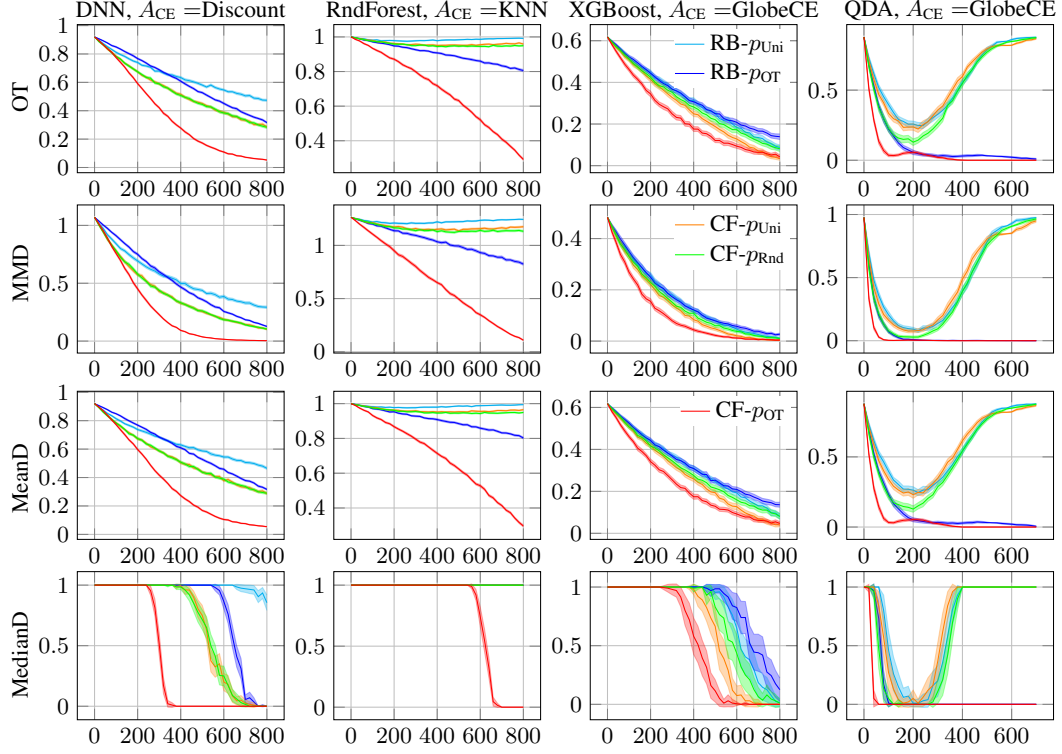


Figure 5: [HELOC] Divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C . Each curve aggregates 100 runs, and the shaded band denotes the 99.9% confidence interval. The legends apply to all panels; the value module is $A_{\text{Value}}^{\text{avg}}$.

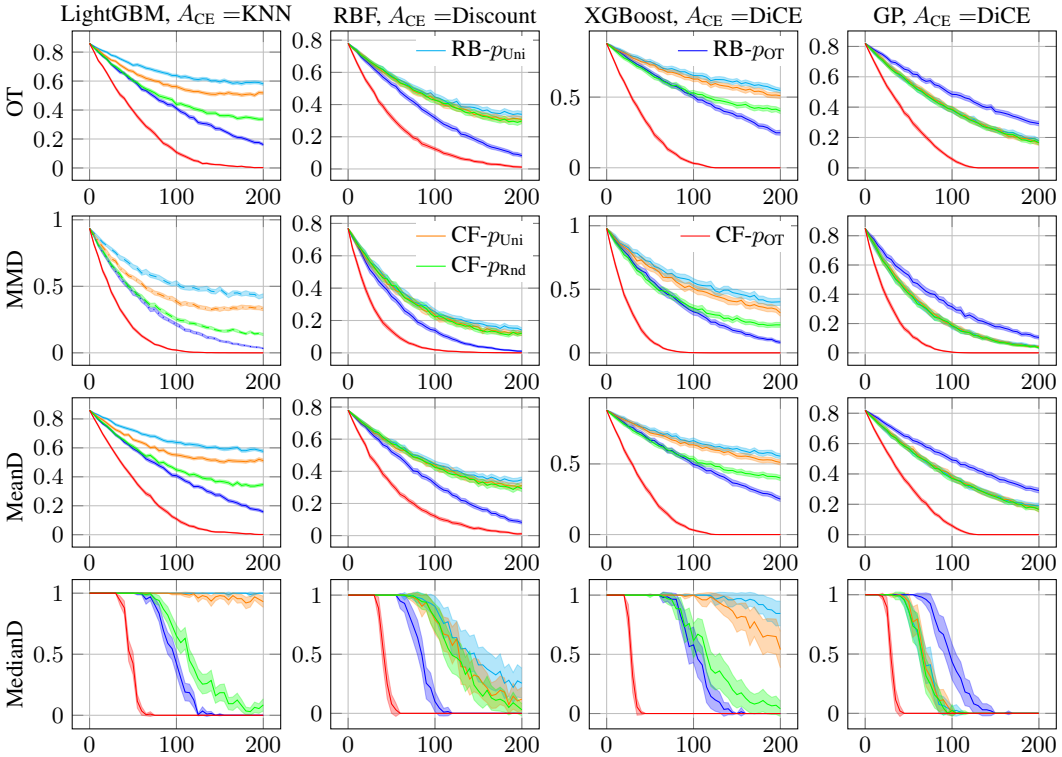


Figure 6: [German Credit] Divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C . Each curve aggregates 100 runs, and the shaded band denotes the 99.9% confidence interval. The legends apply to all panels; the value module is $A_{\text{Value}}^{\text{avg}}$.

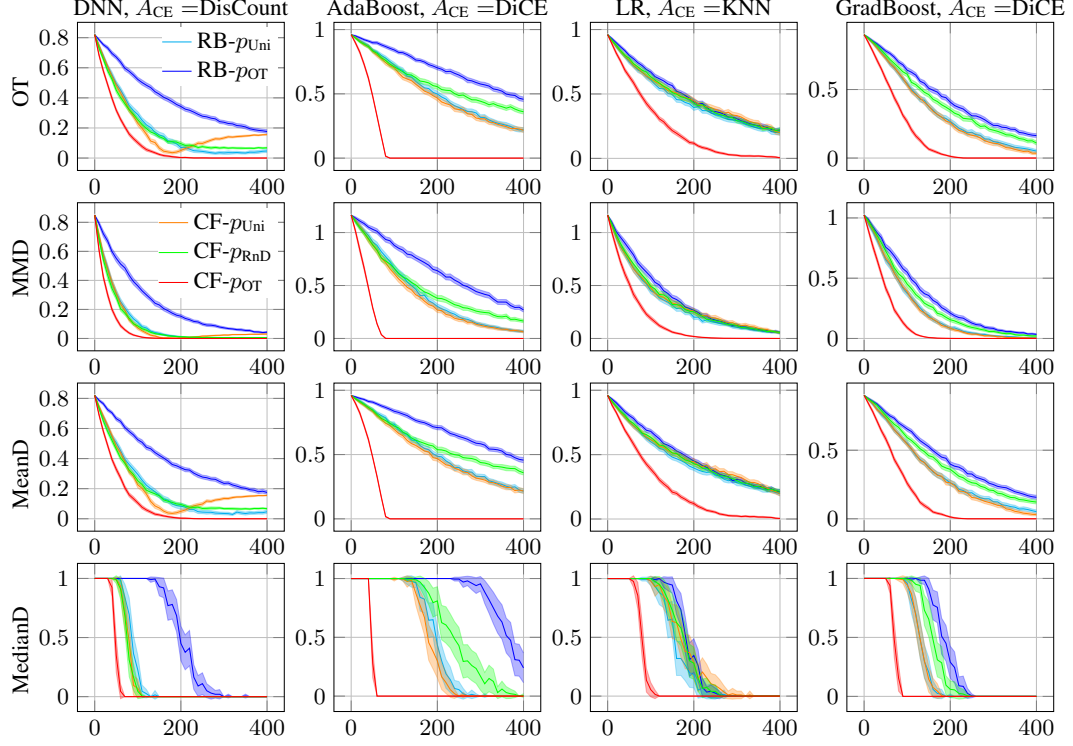


Figure 7: *[Hotel Bookings]* Divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C . Each curve aggregates 100 runs, and the shaded band denotes the 99.9% confidence interval. The legends apply to all panels; the value module is $A_{\text{Value}}^{\max}$.

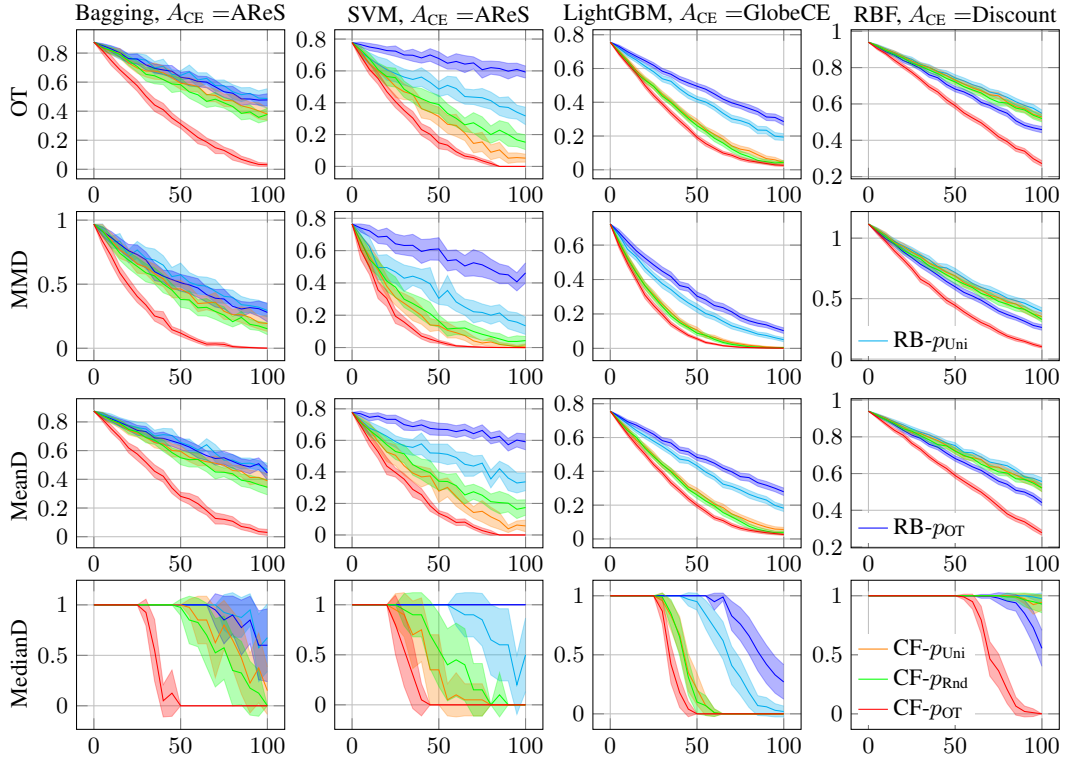


Figure 8: *[COMPAS]* Divergence $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the action budget C . Each curve aggregates 100 runs, and the shaded band denotes the 99.9% confidence interval. The legends apply to all panels; the value module is $A_{\text{Value}}^{\max}$.